

Risk and protective factors of depression among caregivers of children with disabilities in rural Rwanda

Alain Ahishakiye ¹, Leila Dal Santo,² Mary C Smith Fawzi,³ Ann C Miller ³, Matias Placencio-Castro ^{2,4}, Hannah Gilbert,³ Sarah K G Jensen,^{5,6} Kim Wilson,⁵ Catherine M Kirk ⁷, Vincent Sezibera,¹ Theresa S Betancourt^{3,8}

To cite: Ahishakiye A, Dal Santo L, Smith Fawzi MC, *et al.* Risk and protective factors of depression among caregivers of children with disabilities in rural Rwanda. *BMJ Glob Health* 2026;**11**:e020491. doi:10.1136/bmjgh-2025-020491

Handling editor Henry E E Rice

► Additional supplemental material is published online only. To view, please visit the journal online (<https://doi.org/10.1136/bmjgh-2025-020491>).

Received 19 May 2025
Accepted 26 November 2025

ABSTRACT

Background Caregivers of children with disabilities are at increased risk of experiencing poor mental health due to the compound effects of financial and family stressors. This study aims to identify risk and protective factors for poor mental health, particularly depression, among caregivers of young children with disabilities.

Methods This is a secondary data analysis, including a total of 1518 caregivers who were enrolled at baseline in a cluster randomised trial evaluating the effects of Sugira Muryango, a home-visiting parenting intervention, on early child development and violence prevention outcomes. A total of 309 primary caregivers were identified as having a child aged 12–36 months who screened positive for some form of developmental delay. Associations between key sociodemographic and household characteristics and depression outcomes were explored using multivariable linear regression models.

Results Among the sample of caregivers of children with disabilities, 65% met the clinical cut-off for likely depression. Economic hardship ($\beta=0.158$, $p<0.001$), food insecurity ($\beta=0.009$, $p<0.01$) and caregiver illness ($\beta=0.183$, $p<0.01$) were significant risk factors for depression. In contrast, family unity ($\beta=-0.013$, $p<0.001$) was an important protective factor for depression among these caregivers. Among married/cohabiting caregivers, intimate partner violence (IPV) was associated with higher mean depression scores ($\beta=0.306$, $p<0.001$), though the inclusion of family unity in the full model (inclusive of risk and protective factors) attenuated the impact of IPV on depression.

Conclusions Programmes targeting children with disabilities should also address the mental health needs of parents. Mental health difficulties are common for caregivers living in poverty, especially those raising a child with a disability or developmental delay. Additionally, parenting programmes that reduce family violence and build family unity should be prioritised by policymakers interested in improving the mental health and well-being of families facing extreme poverty.

INTRODUCTION

Children born with low birth weight, premature or with neurodevelopmental disabilities in low-resource settings face challenges with

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Caregivers of children with disabilities are at high risk of depression.

WHAT THIS STUDY ADDS

⇒ Food insecurity and caregivers' illness, among others, were the main risk factors of depression among our study participants. Family unity is an important protective factor against depression among caregivers of children with disabilities in rural Rwanda.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Parenting programmes should also promote economic empowerment, reduce family conflict and strengthen family unity.

respect to their development. Such challenges are attributable to a variety of factors, including poor nutritional intake (and consequently, stunting); poverty, lack of stimulation from caregivers and any underlying neurodevelopmental risks.¹ The 2019–2020 Rwandan Demographic and Health Survey report revealed that 14% of children aged 5 or above have a disability in at least one functional domain.² Children with disabilities need additional support beyond their everyday basic needs, such as specialised healthcare and inclusive early learning services.³ In rural Rwanda, many people live with limited resources and may not be able to respond to all these needs.

Given caregivers' vital role in supporting their child's developmental needs, caregiver mental health during early development is crucial for the child's overall development.⁴ Indeed, poor parental mental health can compromise nurturing care practices, which in turn may result in poor childhood development outcomes for children who depend on caregivers for care and age-appropriate stimulation.⁵ Maternal depression, for example, can have a profound impact on childhood



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For numbered affiliations see end of article.

Correspondence to
Dr Alain Ahishakiye;
alanahish@gmail.com

development with a risk of intergenerational transmission of depression, especially in low-resource settings.^{6–8} In Rwanda, parents are already at increased risk of poor mental outcomes because of the legacy of post-colonial economic impoverishment, exacerbated by the 1994 genocide, which greatly affected caregivers' mental status and caused resource scarcity.⁹ Within this historical context, caring for a disabled child can be an added stressor for caregivers, as it often requires additional resources that may be challenging for parents to access in low-resource settings.¹⁰ Indeed, in rural Rwanda, mothers of children with a disability were twice as likely to experience poor mental health compared with those without a child with a disability.¹¹

A host of risk factors for depression among caregivers residing in low-income and middle-income countries (LMICs) has been identified, including poverty and economic stress, domestic violence, low levels of parental education and single marital status.^{12–14} However, few studies have assessed barriers to and facilitators of mental well-being among parents of children with disabilities. The few studies that have explored this topic in LMICs have also identified risk factors consistent with those of the wider population, such as economic hardship and lower levels of education, while highlighting the role of other factors such as parental age and perceived caregiver burden. Studies exploring factors associated with psychological distress among families of children with disabilities in Jordan, Malawi and Nepal, for example, have found that caregiver older age, socioeconomic and employment status, social support, having a female child and social and instrumental support were associated with lower psychological distress scores.^{15–17}

Importantly, these few studies that address the intersection of a child's disability and caregiver mental health in LMICs tend to focus primarily on the challenges of caring for a child with a disability without addressing potential protective factors and available family or community resources that can be leveraged to support caregiver mental health.¹⁸ This research question is critical, as designing an intervention to support caregiver mental health requires attending to the barriers as well as facilitators of mental well-being, particularly in contexts of adversity.¹⁹ This study aims to fill this gap by assessing the mental health status of caregivers of children with disabilities and investigating risk and protective factors associated with mental health outcomes of rural caregivers in three districts of Rwanda. Specifically, we seek to identify factors that predict or protect against poor mental outcomes among caregivers of children with developmental disabilities in rural Rwanda.

METHODS

Study population

Data for this secondary analysis come from the baseline assessment of a cluster randomised trial (CRT) designed to test the effects of Sugira Muryango, a home-visiting

parenting intervention, on early child development and violence prevention outcomes. The CRT was conducted in Nyanza, Ngoma and Rubavu districts in Rwanda, selected due to the presence of the Rwandan government's social protection programme, 'Vision 2020 Umurenge Programme' (VUP), and the desire to minimise overlap with other governmental and non-governmental early childhood development (ECD) programmes. The VUP, established in 2007, offers direct support (eg, unconditional cash transfers), nutrition-sensitive direct support and public works programming to families living in extreme poverty (Ubudehe 1, the most extreme poverty level). The public works component of VUP consists of two cash-for-work schemes: classic public works, which provides cash for short-term, temporary employment on community infrastructure projects, and an expanded public works programme, characterised by more flexible (part-time, year-round) and less strenuous labour for cash. Details regarding the cluster randomisation procedure for the CRT have been reported elsewhere.²⁰ In brief, randomisation was done at the cluster level, stratified by public works type (expanded public works only, combined expanded works/classic public works, and classic public works only). A total of 1049 VUP-eligible families with children aged 6 to 36 months were randomised to receive either treatment (n=508 families) or usual care (n=541 families).

Participants for this analysis include primary caregivers of children with disabilities who participated in the baseline assessment in May 2018. The primary caregiver was defined as the primary provider of care for the child with disabilities—this person could be the biological mother, father or a close family member. At baseline, 1518 caregivers (from 1049 eligible households) were enrolled in the trial, of whom 1066 were primary caregivers. A total of 309 caregivers were included in the final sample, among whom 149 were married or cohabiting, because their children aged 12–36 months screened positive for some form of developmental delay. Child disability was assessed using seven items adapted from the UNICEF 10-item screener, Multiple Indicator Cluster Survey version,²¹ a widely used tool to identify children with developmental delays or at risk of disability. A child is screened as positive for having a disability if the caregiver answers yes to one or more items.

Figure 1 presents a flow chart depicting the process of identifying eligible participants for this analysis.

Description of the intervention

The Sugira Muryango home-visiting programme aims to improve ECD and reduce family violence through active coaching in responsive parenting, strengthening problem-solving skills and social support, and building coping and parental emotion regulation skills by involving both female and male caregivers. The intervention targeted families at the bottom rank in the Ubudehe classification scheme who had one or more children

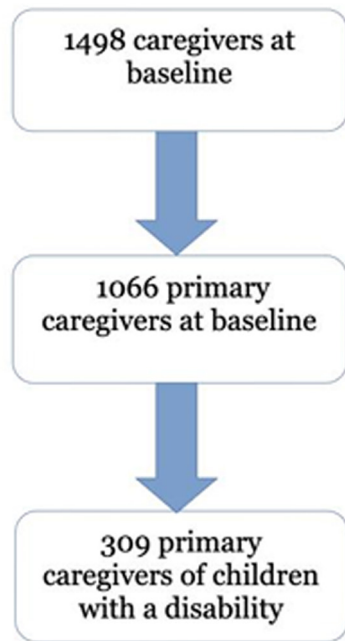


Figure 1 The process of identifying eligible participants for this analysis.

aged 6–36 months and were willing to participate in the parenting intervention.²²

Patient and public involvement

While patient and public involvement was not integral to this secondary data analysis, the Sugira Muryango manual and materials were developed with input from local and international ECD experts, Rwandan government partners and community advisory boards.

Outcome measures

Depression

The Hopkins Symptoms Checklist (HSCL-25), a screening tool used to measure symptoms of anxiety and depression, was used to assess depressive symptoms among the sample. The HSCL-25 comprises a 10-item scale for evaluating anxiety symptoms and a 15-item subscale for depression symptoms—the latter scale was used for this study. Each item is rated on a Likert scale ranging from 1 (not at all) to 4 (extremely), reflecting the intensity of the symptoms experienced over the past month. The HSCL-25 for depression subscale, previously validated for use in Rwanda, demonstrated strong internal consistency among the sample ($\alpha=0.87$).²³ Consistent with the literature, we used a cut-off of 1.75 on the subscale to indicate clinically relevant depressive symptoms among caregivers of children with disabilities.²⁴

Parental education

A binary variable assessed caregiver education, whereby a value of 0 corresponded with no education up to primary education (6 years) and a value of 1 corresponded with having a primary school level of education or higher.

Family unity

The family unity screening tool is a locally derived, 15-item scale used to assess the level of *kwinzerana* (family trust and unity) among children, adolescents and caregivers in Rwanda.^{25 26} The scale was first developed following qualitative research on mental health problems and resilience among children and families affected by HIV in Rwanda; this research identified *Kwizerana* as an important protective factor for children affected by HIV/AIDS.^{25 27} Each item from the assessment is scaled between 0 and 4 points (0 corresponding to ‘never’ and 4 corresponding to ‘always’), with higher scores indicative of higher levels of family unity. Example items from the questionnaire, which covers facets of family functioning such as emotional connectedness, peaceful coexistence, trust and mutual support, include the following: ‘we interact with each other/we socialise between us’, ‘we converse with each other/we talk to reach agreements’, ‘we understand each other’, and ‘we live together in peace’. A total family unity score was calculated by summing responses to these 15 items. The family unity scale demonstrated strong reliability among this sample, with a Cronbach’s alpha of 0.96.²⁸

Intimate partner violence

The Conflict Tactics Scales^{29 30} were used to measure various types of intimate partner violence (IPV), including physical, emotional and sexual abuse. We created a binary variable to assess IPV victimisation, where respondents were coded as one if they reported being the victim of any type of IPV and 0 if they did not.

Economic hardship

Economic hardship in the past 6 months was assessed using an adapted version of the Post-War Adversities Index, a 14-item self-report inventory of stressful life events and circumstances related to interpersonal adversities, hunger, and housing and economic insecurity during the last 6 months.³¹ A total economic hardship variable was generated based on the sum of six dichotomously scored items that assessed housing and economic insecurity, including ‘living accommodations inadequate or unfit to live in the last 6 months,’ ‘lacked money for necessities’ and ‘the primary wage earner was unemployed’.

Food insecurity

The Coping Strategy Index (CSI), a tool used to measure the frequency and intensity of coping strategies for households reporting a shortfall in food for consumption, was used to assess food insecurity among the study sample.³² Participants were asked to report how many days in the week preceding the interview they used each coping strategy. Examples include ‘relying on or less preferred and less expensive foods,’ ‘borrowing food,’ and ‘limiting portion size at mealtimes.’ Per the scoring guidance outlined in the Rwanda 2015 Comprehensive Food Security and Vulnerability Analysis Report,³³ a total

CSI score was computed as the sum of weighted scores, with higher CSI scores indicating higher levels of food insecurity. Next, we categorised coping into four categories, including 'no coping' (CSI score=0), 'low coping' (CSI scores between 1 and 10), 'medium coping' (CSI scores between 11 and 18) and 'high coping' (CSI scores over 18).

Statistical analysis

Descriptive analyses, including percentage distributions, means and frequency distributions, were conducted for all sociodemographic variables. Less than 1% of respondents had missing data across all variables included in this analysis, so no additional steps were taken to address the missingness. All explanatory variables were then included in bivariate analyses. Specifically, we performed paired t-tests and one-way analysis of variance tests to explore bivariate associations between categorical variables and the outcome. Pearson correlation tests assess this relationship for continuous explanatory variables. We also explored associations between sociodemographic variables and risk and protective factors separately so as to avoid issues of collinearity. Finally, explanatory variables found to be marginally significant ($p < 0.10$) in bivariate analyses were tested in multivariable linear regression Ordinary Least Squares (OLS) models with cluster-robust SEs to account for clustering. Regardless of the significance in the bivariate analyses, age was included in these regression models. Three main-effects regression models were fitted to assess the impact of including successive groups of covariates on mean depression scores: Sociodemographic variables were included in the first model, followed by risk factors in the second model, and protective factors in the final model. Parameter estimates (coefficients), model fits and differences in R^2 are reported for each regression model. Additional analyses among married/cohabiting-only participants were conducted to determine the contribution of predictors such as IPV on caregiver depression. All quantitative analyses were conducted using Stata V.18.1. The study findings are reported according to the Strengthening the Reporting of Observational Studies in Epidemiology checklist.³⁴

RESULTS

Depression scores were high among the study sample, with the majority (65%) of caregivers exceeding the 1.75 threshold suggestive of probable depression (table 1). In terms of sociodemographic characteristics, the mean age of the study participants was 34 years. Just over half (52%) were single, non-married parents. Similar distributions were observed for dual caregiver household status: The majority (54%) of caregivers cited being in a single-caregiver household. The overwhelming majority (88%) of the caregivers were mothers. Three-quarters of participants had low levels (less than 6 years) of education (75%). Education levels among caregivers were similar across all three districts.

Table 1 Selected characteristics of primary caregivers of children with disabilities in rural Rwanda (n=309)

Characteristic	N	(%) or mean (SD)
Sociodemographic variables		
Mean age	309	33.9 (9.3)
Sex of the primary caregiver		
Male	5	1.6
Female	304	98.4
Marital status		
Single/separated/divorced/widowed	160	51.8
Married/cohabiting	149	48.2
Dual caregiver household (at enrolment)		
No	168	54.4
Yes	141	45.6
Caregiver relationship with the child		
Mother	273	88.3
Father	5	1.6
Aunt/uncle/grandparent/other	31	10.0
Educational attainment		
Less than 6 years of primary	233	75.4
6 years primary or higher	76	24.6
District		
Ngoma	98	31.7
Nyanza	111	35.9
Rubavu	100	32.4
Risk factors		
Caregiver has disability		
No	275	89.0
Yes	34	11.0
Serious illness in the past 12 months (caregiver)		
No	178	57.6
Yes	131	42.4
The mean number of children under 5 years of age in the household	309	1.5 (0.6)
The mean number of economic hardships in the past 6 months	308	1.85 (1.0)
Any daily hardship in the past 6 months		
No	16	5.2
Yes	292	94.8
Household food insecurity (less expensive food, borrowed food, limit food portion, limit food consumption, reduce meals) high score is worse)		
No coping	7	2.3
Low coping	35	11.3
Medium coping	67	21.7
High coping	200	64.7
Mean household food insecurity score	309	23.1 (11.4)

Continued

Table 1 Continued

Characteristic	N	(%) or mean (SD)
Victim of any abuse (among married/cohabiting participants)		
No	68	46.3
Yes	79	53.7
Protective factors		
Health insurance (all members of the family)		
No	118	38.2
Yes	191	61.8
Public works type		
Classic	214	69.3
Expanded	95	30.7
Mean family unity score	309	47.3 (13.6)
Study outcome		
Meets threshold for depression (HSCL-25 score ≥ 1.75)	309	65.0
HSCL-25, Hopkins Symptoms Checklist.		

Regarding risk factors for depression, almost half of the participants experienced severe illness in the past 12 months (42%). Nearly all participants (95%) experienced daily economic hardship in the past 6 months, and nearly two-thirds (65%) cited using high coping strategies to deal with food insecurity. Among married participants, slightly over half (54%) reported experiencing emotional, physical or sexual IPV.

Concerning protective factors, 62% of participants indicated that all family members were covered by health insurance, and 69% received cash support from the classic vs expanded public works programme.

Results from bivariate analyses between mean depression scores and each theorised risk and protective factor are presented in [table 2](#). Correlations between each sociodemographic, risk and protective factor are available in online supplemental appendix tables 1 and 2. Significant differences in mean depression score were observed for married/cohabiting status, dual caregiver status and caregiver level of education. Specifically, mean depression scores were significantly higher among single caregivers compared with caregivers' dual-caregiver households ($p<0.01$). Similar patterns were observed for married or cohabiting caregivers compared with their non-married or cohabiting counterparts ($p<0.05$). In addition, caregivers with more than 6 years of education had a lower mean depression score compared with caregivers with six or fewer years of education ($p=0.04$) ([table 2](#)).

Significant associations at the $p<0.1$ level were observed between depression and the following risk factors: IPV victimisation (any abuse), economic daily hardship, household food insecurity and caregiver illness in the past 12 months. Caregivers who cited IPV victimisation of any kind reported higher mean depression values compared with those who did not experience IPV victimisation. Similarly, mean depression scores among

Table 2 Differences in mean depression score by sociodemographic characteristics and risk and protective factors among caregivers of children with a disability in Rwanda (n=309)

Variable	Mean (SD)	F/t	P value
Sociodemographic			
Sex			
Male	1.99 (0.68)	t=−0.12	0.90
Female	2.02 (0.63)		
Married/cohabiting			
No	2.09 (0.59)	t=2.06	0.04
Yes	1.94 (0.66)		
Dual caregiver HH			
No	2.11 (0.59)	t=2.84	0.005
Yes	1.91 (0.66)		
Caregiver relationship with the child			
Mother	2.00 (0.63)	t=1.56	0.12
Father/other	2.18 (0.61)		
Education			
<6 years primary	2.06 (0.61)	t=2.07	0.04
6 years primary or higher	1.89 (0.68)		
District			
Ngoma	2.04 (0.63)	F=0.06	0.94
Nyanza	2.02 (0.64)		
Rubavu	2.01 (0.62)		
Risk factors			
Caregiver disability			
No	2.02 (0.64)	t=−0.40	0.69
Yes	2.06 (0.61)		
Victims of any IPV abuse			
No	1.73 (0.63)	t=−3.57	<0.001
Yes	2.11 (0.65)		
Caregiver illness			
No	1.87 (0.60)	t=−5.07	<0.001
Yes	2.23 (0.62)		
Protective factors			
Insurance for all family members			
No	2.05 (0.64)	t=0.68	0.49
Yes	2.00 (0.63)		
Public works type			
Classic	1.98 (0.62)	t=−1.81	0.07
Expanded	2.12 (0.65)		
HH, household; IPV, intimate partner violence.			

HH, household; IPV, intimate partner violence.

caregivers who cited experiencing a serious illness in the past 12 months were significantly higher than for those who had not experienced a major illness ($p<0.001$). Finally, as shown in the correlation matrix in (online supplemental appendix table 1), significant correlations ($p<0.001$) were observed between the mean depression

Table 3 Coefficients and 95% CIs from multivariable regression analyses examining sociodemographic characteristics and risk and protective factors associated with depression among caregivers of children with a disability in Rwanda (n=309)

Variables	Full sample (n=309)			Married/cohabiting only (n=149)		
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
Step 1 (sociodemographics)						
Age	0.003 (−0.006 to 0.013)	0.003 (−0.005 to 0.011)	0.003 (−0.005 to 0.011)	0.006 (−0.011 to 0.023)	0.010 (−0.003 to 0.024)	0.005 (−0.008 to 0.019)
Dual caregiver HH	−0.191 (−0.337 to −0.045)*	−0.180 (−0.307 to −0.052)**	−0.174 (−0.304 to −0.045)**	−0.409 (−0.748 to −0.069)*	−0.268 (−0.593 to −0.057)	−0.358 (−0.684 to −0.033)*
Relationship to child	−0.097 (−0.366 to 0.171)	−0.079 (−0.330 to 0.172)	−0.118 (−0.350 to 0.113)	0.073 (−0.334 to 0.482)	0.069 (−0.271 to 0.411)	−0.014 (−0.321 to 0.291)
Education level	−0.170 (−0.344 to 0.004)	−0.177 (−0.330 to −0.024)*	−0.160 (−0.304 to −0.016)*	−0.135 (−0.384 to 0.114)	−0.130 (−0.333 to 0.073)	−0.061 (−0.262 to 0.138)
Step 2 (risk factors)						
Mean economic daily hardships		0.172 (0.103 to 0.242)***	0.158 (0.089 to 0.228)***		0.205 (0.099 to 0.311)***	0.190 (0.077 to 0.303)**
Mean HH food insecurity score		0.009 (0.003 to 0.015)**	0.009 (0.003 to 0.015)**		0.008 (−0.001 to 0.018)	0.006 (−0.003 to 0.016)
The caregiver had a serious illness in the past year		0.196 (0.053 to 0.340)**	0.183 (0.046 to 0.320)**		0.290 (0.084 to 0.496)**	0.273 (0.079 to 0.467)**
Ever a victim of IPV abuse	--	--	--		0.306 (0.104 to 0.509)**	0.153 (−0.037 to 0.345)
Step 3 (protective factors)						
Family unity score			−0.013 (−0.018 to −0.008)***			−0.014 (−0.023 to −0.005)**
Public works type			0.025 (−0.118 to 0.169)			−0.017 (−0.267 to 0.232)
Model R ² (Δ R ²)	0.05	0.24 (0.19)	0.33 (0.09)	0.04	0.36 (0.32)	0.42 (0.06)
Overall F-test model fit	Prob>F=0.016 F(4,152)	Prob>F<0.001 F(7,152)	Prob>F<0.001 F(9,152)	Prob>F=0.051 F(4,101)	Prob>F<0.001 F(8,100)	Prob>F<0.001 F(10,100)

***: P<0.001; **: P<0.01; *: P<0.05.

HH, household; IPV, intimate partner violence.

score, mean number of economic hardships and mean food insecurity score, with Pearson correlation coefficients of 0.37 and 0.30 for economic hardship and food insecurity, respectively.

Two protective factors emerged as significant at the p<0.10 level in bivariate analyses. Specifically, caregivers in households who participated in an expanded public works programme had slightly lower mean depression scores than their classic public works counterparts (p=0.07) (table 2). Next, a significant negative association was observed between mean family unity and depression scores, with a Pearson correlation coefficient of −0.32 (p<0.001) (online supplemental appendix tables 1).

Table 3 displays results from the multiple regression models examining the role of sociodemographic, risk and protective factors on depression scores. Concerning sociodemographic factors, age was not significantly associated with depression scores in any of the fitted models for the full sample and married/cohabiting sample. Dual caregiving was significantly associated with lower depression scores across all models for the full sample, though the magnitude of this effect changed slightly with the addition of risk and protective factors (model

1: β=−0.190, p<0.05; model 2: β=−0.180, p<0.01; model 3: β=−0.174, p<0.01). Dual caregiving status was also significantly associated with lower depression scores among married/cohabiting participants, with a coefficient of −0.358 (p<0.05) in the final model. Across all models for both the full and married/cohabiting samples, no significant relationship was observed with respect to the caregiver being the mother, the father or the aunt/grandparents. Higher levels of education were significantly associated with lower mean depression scores among the full sample in models 2 and 3 (p<0.05). Among married participants, educational attainment had no impact on depression scores across all models.

In terms of risk factors for depression, experiencing economic hardship was significantly associated with higher depression scores across models (model 2: β=0.172, p<0.001; model 3: β=0.158, p<0.001). Among married respondents, economic hardship was also significantly associated with increased mean depression scores in models 2 and 3, with a coefficient of 0.205 (p<0.01) and 0.190 (p<0.001) in the second and final fitted models, respectively. Small, positive associations were observed between food insecurity scores and mean depression scores in the second and

third models, though this relationship was not observed for married participants. Being seriously ill in the past year was significantly associated with higher depression scores ($\beta=0.183$, $p<0.01$). Even greater coefficients were observed for this relationship among married respondents, where severe illness in the past year was associated with a 0.27-point increase ($p<0.01$) in the caregiver's mean depression score in the final model. For married respondents, experiencing IPV was one of the strongest risk factors for depression among caregivers of children with disabilities: In model 2, IPV victimisation was associated with a 0.31-point increase in mean depression score, though this association was no longer significant when adding family unity in the final model.

Finally, in terms of the role of protective factors on caregiver depressive symptoms, family unity emerged as a significant protective factor, associated with lower levels of depression for both the full sample ($\beta=-0.013$, $p<0.001$) and married/cohabiting respondents ($\beta=-0.014$, $p<0.01$) in the final models. Greater family cohesion was associated with lower levels of depression, indicating the importance of strong family cohesion. A higher level of education was negatively associated with depression scores ($\beta=-0.160$, $p<0.01$ in the final model). However, the contribution of at least a primary level of education to the reduction in depression score was not significant among married couples.

DISCUSSION

This study aimed to illuminate risk and protective factors for depression among caregivers of children with disabilities in rural Rwanda. Prior to reflecting on the results from bivariate and multivariable analyses, it is worth noting the high levels of depression among this sample of caregivers. Nearly two-thirds (65%) of respondents in this sample had a score of 1.75 or higher on the depression subscale of the HSCL, and a cut-off score of 1.75 is traditionally considered to be indicative of probable clinical depression.^{35–37} This high prevalence rate is consistent with—though still higher than—recent studies conducted across several countries in sub-Saharan Africa, which found a caregiver depression prevalence ranging from 41% to 52% and around 40%.^{11 16 38} As Rwanda works to implement its National Mental Health Policy,³⁹ these findings underscore the importance of addressing mental healthcare gaps among certain subgroups of the population, such as caregivers of children with disabilities.

Single-caregiver versus dual-caregiver household status and education level emerged as important sociodemographic predictors of depression among this sample, findings consistent with the wider literature on risk factors for depression among caregivers residing in LMIC.^{12 14} Moreover, two studies recently conducted in Ethiopia found similar results with respect to single-caregiver status, how carrying the burden of poverty, stigma and lack of understanding from others related to having a child with a disability as a single parent was an important

determinant of their quality of life.^{40 41} Regarding education as a protective factor, our finding that higher levels of education were protective against depressive symptoms (though not among married respondents) is consistent with the research literature on mental health problems among caregivers of children with a disability in LMICs. For example, illiteracy was identified as an important risk factor for psychological distress in a study of psychological distress among rural Nepali caregivers of children with a neurodevelopmental disorder.¹⁷ For our study population, it could be the case that caregivers with greater education have more knowledge of their child's respective disability and are therefore more confident in managing the disability. In addition, as education is widely recognised as an important predictor of care-seeking behaviour, caregivers with higher levels of education may be better able to access information, find relevant resources and navigate informal and formal support for their children with special needs. Finally, higher levels of education are generally associated with better employment opportunities for workers in LMICs, which could translate into reduced caregiver economic-related stress.

Consistent with the extant literature, economic hardship emerged as a salient risk factor for depression in this analysis and was significantly associated with increased depression scores among the full sample of caregivers of children with disabilities, as well as among the married subsample. Curiously, our study revealed that the magnitude of the association between economic hardship and depression was even *greater* for the married/cohabiting subsample. This finding could be explained by married women's lack of decision-making authority with regard to household expenditure. The mental health implications of economic hardship for married female caregivers who do not have any say in how household money is spent may be greater than for single caregivers who are able to make decisions about household expenses. Future analyses are warranted to better understand the mechanisms underlying this relationship among married vs unmarried caregivers. Irrespective of marital status, this study underscores the toll of economic hardship on depression among caregivers of a child with disabilities, which has important policy implications given the added expenses these families likely face in raising a child with a disability.

Household food insecurity was also significantly associated with increased depressive symptoms among caregivers of children with disabilities in rural Rwanda, a finding consistent with previous work on the relationship between food insecurity and caregiver depression in sub-Saharan Africa.^{42 43} Approximately 85% of the study sample cited engaging in medium or high coping strategies to deal with food insecurity. Poor mental health could be explained by the stress related to these coping strategies, including reducing expenses allocated to food, borrowing food from the merchants and limiting portion size at mealtimes. The implications of food insecurity for caregiver mental health (not to mention the possible mental and physical health consequences of

food insecurity for children) are important for policy-makers to consider as they determine how best to allocate resources to impoverished families in Rwanda.

Serious caregiver illness in the past 12 months was another important risk factor for depressive symptoms among our study sample. This finding could be explained by the cumulative stress felt by the caregivers who struggle with illness while simultaneously attending to the needs of their child with a disability. Given the important linkages between poverty and limited access to health-care in sub-Saharan Africa,⁴⁴ it is also possible that the depressive symptoms observed among caregivers in this sample are related to the stress associated with barriers to healthcare access. Having health insurance for every household member was not associated with depressive symptoms in our bivariate analysis, though it could be the case that poor families may not have sufficient funds to pay the additional fees for medication and services that are not covered by health insurance. Further analyses are warranted to understand to what extent out-of-pocket medical expenses, whether for caregivers themselves or their children with disabilities, contribute to mental health difficulties among this population.

Consistent with the literature on the link between IPV and depression, we found that experiencing IPV was associated with increased depression scores among married caregivers of children with disabilities, though this relationship was no longer significant on the inclusion of family unity in the model. Prior studies in Rwanda have explored domestic violence within families and its association with common mental health disorders, notably depression and Post-Traumatic Stress Disorder (PTSD) among caregivers in the post-genocide era^{45–48} and the impact of IPV on offspring's mental health.⁴⁹ This is the first study to examine this relationship among caregivers of children with a disability. As such, our study contributes to and deepens this literature by probing the relationship between IPV experience and caregiver depression among this particular segment of the Rwandan population.

Finally, family unity emerged as an important protective factor against depression among caregivers of children with disabilities in rural Rwanda. Indeed, as noted above, accounting for the experience of IPV in the household, higher family unity remained significantly associated with lower depression scores. Given that family dynamics play a key role in both caregiver and child well-being, this finding is not surprising. Among studies conducted in Rwanda specifically, family unity had previously been identified as a protective factor worth including in future parenting programmes to reduce mental health problems among families affected by HIV/AIDS.²⁵ The introduction of a parenting programme promoting family unity among caregivers affected by HIV in Rwanda, based on those findings, was associated with a reduction of IPV and an improvement in family functioning.⁵⁰ More recent research on the Sugira Muryango parenting programme, which aims to strengthen family unity, was also found to improve mental health outcomes of caregivers living in extreme poverty in

rural Rwanda.²⁰ Findings from our study corroborate this body of research, underscoring the importance of family unity for mental health in contexts of adversity.

Our study has several implications for policymakers working at the intersection of health and social protection. While the Rwandan Social Protection Program (VUP) has markedly improved the lives of the country's most vulnerable population,^{51 52} this study highlights the burden of depression among caregivers of children with disabilities and identifies relevant risk and protective factors to better support this population. From a social protection standpoint, interventions to reduce the range of economic hardships and food insecurity facing these caregivers are imperative and could include targeted cash transfers, expanded health insurance coverage and support for income-generating activities/livelihood training.⁵³ Moreover, given that ECD parenting programmes in LMICs (and Rwanda specifically) have shown great promise in improving child health as well as caregiver mental health outcomes,^{20 22 54 55} it will be important to consider how these programmes can be tailored to provide additional support for female caregivers of children with disabilities.^{20 56 57}

Limitations

Several limitations of the present study are worth noting. First, given the use of secondary data for our study, we were limited to the data available to us. As such, we could not explore other potentially relevant covariates of maternal depression that have previously been identified in the research literature, such as social support, perceived caregiver burden and stigma related to child disability. Next, we chose to limit our study sample to primary caregivers of children with a disability, given the dearth of research on the mental health needs of this group, particularly in Rwanda. Thus, we are unable to know about the mental well-being of other caregivers in the household, caregivers who might not be caring for the child during the day, but who might still be working to support this child and invest in his/her well-being. Future work on this topic could expand the study sample to include all caregivers living in the household. Regarding study design, we acknowledge that our use of cross-sectional data limits our ability to establish causal inferences regarding the relationship between hypothesised covariates and caregiver depressive symptoms. For example, it could be the case that certain economic hardships, such as unemployment, are due to poor mental health. A longitudinal study design would be better suited to explore causal links between certain socio-economic conditions (eg, poverty) and depression. In addition, we would have liked to explore any differences in risk and protective factors between male and female caregivers in our study, particularly given the high percentage of female, single caregiver households in our sample. Unfortunately, the very small number of male participants (less than 2% of the sample—five participants) precluded any additional sex-disaggregated analyses. Future studies that benefit from a larger sample size of male caregivers should consider investigating differences in depression

scores by sex as such information would help inform policy targeting and gender-oriented interventions. Finally, while quantitative analyses are an important starting point for understanding the mental health burden (and relevant risk and protective factors) among caregivers of children with a disability in Rwanda, this approach prevents us from contextualising findings in the way that qualitative research would allow. Future qualitative studies are warranted to provide a more in-depth understanding of the experience of depression among our study population. Despite these limitations, we believe this study represents an important contribution to the literature on the mental health needs of caregivers with disabilities, with findings that are generalisable to those with similar sociodemographic characteristics, namely caregivers of children with disabilities living in extreme poverty and rural regions of LMICs.

CONCLUSIONS

This study aimed to examine the correlates of depression among caregivers of children with a disability in rural Rwanda. Our findings revealed significant associations between depressive symptoms and a host of individual- and household-level risk and protective factors, including parental education, economic insecurity, caregiver illness and family unity, among others. These results have important implications for policymakers and researchers alike, offering insight into the types of stressors that could be alleviated through social protection, ECD-focused parenting programmes and mental health services. National programmes that focus on poverty alleviation, child development and family well-being, such as the VUP and Rwanda's Mental Health Policy, should consider integrating caregiver mental health support into their existing programming.

Author affiliations

¹Centre for Mental Health, University of Rwanda, Kigali, Rwanda

²Social Work, Boston College University, Boston, Massachusetts, USA

³Global Health and Social Medicine, Harvard Medical School, Boston, Massachusetts, USA

⁴Lynch School of Education and Human Development, Boston College, Chestnut Hill, Massachusetts, USA

⁵Boston Children's Hospital, Boston, Massachusetts, USA

⁶Harvard Medical School, Boston, Massachusetts, USA

⁷Independent Researcher, Partners In Health/Inshuti Mu Buzima, Rwinkwavu, Rwanda

⁸Research Program on Children and Adversity/School of Social Work, Boston College University, Boston, Massachusetts, USA

Contributors All authors contributed to the study's conception and design. AA, TSB, ACM, MCSF, LDS and MP-C led data analysis, while AA and LDS, ACM, TSB, MCSF, HG, VS, KW, CMK and SKGJ analysed and interpreted the data. AA drafted the initial manuscript, and all authors critically revised it for important intellectual content. All authors approved the final version of the manuscript. AA is the guarantor of this work and accepts full responsibility for the integrity of the data and the accuracy of the analysis.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Disclaimer The findings, interpretations and conclusions stated in this paper are entirely those of the authors.

Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Patient consent for publication Not applicable.

Ethics approval This study used the public use file from the Sugira Murungu Early Childhood Development Home Visiting Intervention Trial (2017–2019), which is available on the World Bank Microdata Library.⁵⁸ The study did not involve human subjects. Only de-identified data were used for this secondary data analysis.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available in a public, open access repository.

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ORCID iDs

Alain Ahishakiye <https://orcid.org/0000-0002-5007-226X>

Ann C Miller <https://orcid.org/0000-0001-6841-9439>

Matias Placencia-Castro <https://orcid.org/0000-0003-4927-3344>

Catherine M Kirk <https://orcid.org/0000-0001-9103-2611>

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